Problem-Solving Insights III

The Grand Calculus of the Universe

Lexington Physics Club

All human measurements are made over finite intervals, which means, we all live in the grand calculus of the universe...

This handout consists of 11 tricky problems that focus on the applications of calculus in physics.

1. A simple pendulum has length L and undergoes small oscillations in a uniform gravitational field of strength g. Its period is
$$T = 2\pi\sqrt{\frac{L}{g}}.$$
   (a) The length is increased by 3%, while g is increased by 1%. Estimate the percentage change in the period without evaluating any square roots.

   (b) More generally, let the fractional changes in L and g be ϵ and η, respectively. Find the fractional change in T to first order in these quantities.

   (c) Extend your result to second order. Include the term proportional to $\epsilon\eta$.

   (d) Suppose L and g are increased by the same fraction. Determine whether the period is unchanged only approximately or exactly.

   (e) A pendulum clock is initially accurate. Its length is then increased by 0.2%, with g unchanged. Estimate how much time the clock loses during one day of actual elapsed time.

2. A particle of mass m moves along the x axis under a force $F(x)$. Its position and velocity are differentiable functions of time.

   (a) Use the chain rule to show that, on an interval where velocity can be regarded as a differentiable function of position,
$$a = v\frac{dv}{dx}.$$
   (b) Starting from Newton's second law, derive
$$\frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2 = \int_{x_1}^{x_2} F(x)\,dx.$$
   (c) A particle starts from rest at $x = 0$ and is subject to
$$F(x) = F_0\left(1 - \frac{x}{L}\right),$$

where F_0 and L are positive constants. Find its speed as a function of position during its initial motion to the right.

(d) Find the position at which the speed is greatest and the position at which the particle first turns around.

(e) Find a potential energy function with $U(0) = 0$. Determine the equilibrium position and whether it is stable.

3. A projectile is launched with speed v_0 from an inclined plane that makes an angle α with the horizontal, where $0 \leq \alpha < \pi/2$. The launch direction makes an angle θ above the plane, with $0 < \theta < \pi/2 - \alpha$. Neglect air resistance.

   (a) Choose horizontal and vertical coordinates with the launch point at the origin. Find the trajectory $y(x)$.

   (b) Find the time at which the projectile next meets the plane.

   (c) Find the distance R along the plane from the launch point to the landing point.

   (d) For fixed v_0 and α, use differentiation to find the launch angle that maximizes R. Find the maximum range.

   (e) Check your result when $\alpha = 0$. As the plane becomes steeper, does the optimal launch direction move closer to the plane or farther from it?

4. A block of mass m rests on a rough horizontal surface. The coefficient of static friction is $\mu > 0$. A force of magnitude F is applied at an angle θ above the horizontal, where $0 \leq \theta < \pi/2$.

   (a) Find the normal force while the block remains in contact with the surface.

   (b) Find the value of F required to bring the block to the verge of sliding.

   (c) Use differentiation to find the pulling angle that minimizes this force, and determine the minimum force. Verify that contact is maintained at this angle.

   (d) The surface is now inclined at an angle α, where $0 < \alpha < \arctan \mu$. The applied force lies in the vertical plane of the slope and makes an angle θ above the uphill direction along the surface. Find the minimum force required to initiate uphill sliding, optimizing over $0 \leq \theta < \pi/2$.

   (e) Under the same restrictions on the applied force, find the minimum force required to initiate downhill sliding. Explain why its optimal direction differs from that for uphill sliding.

5. A projectile is launched from the origin and must pass through a point with coordinates (d, h), where $d > 0$ and $h > 0$. Its initial speed and launch angle are v_0 and θ. Neglect air resistance.

   (a) Find an expression for v_0^2 in terms of d, h, and θ. State the allowed range of θ.

   (b) Set $u = \tan\theta$. Use differentiation to find the smallest launch speed for which the point can be reached, and find the corresponding launch angle.

(c) For $d = 3.0\,\mathrm{m}$, $h = 4.0\,\mathrm{m}$, and $g = 10\,\mathrm{m\,s^{-2}}$, evaluate the minimum speed and the corresponding angle.

(d) Is the projectile at its highest point when it passes through (d, h) at the minimum launch speed? Determine the slope of the trajectory there.

(e) Now fix v_0 and d. Find the greatest height attainable at horizontal coordinate d by varying the launch angle. Hence determine the boundary of the region reachable with this launch speed.

6. A charge $+q$ is placed at $x = 0$, and a charge $-q$ is placed at $x = d$, where $q, d > 0$. For $x > d$, the electric field is

$$E_x(x) = \frac{q}{4\pi\epsilon_0}\left(\frac{1}{x^2} - \frac{1}{(x-d)^2}\right).$$

(a) For $x \gg d$, find the first two nonzero terms in the expansion of $E_x(x)$ in powers of d/x.

(b) Explain why the leading field decreases as x^{-3} rather than x^{-2}.

(c) With the potential chosen to vanish at infinity, write the exact potential $V(x)$. Expand it to obtain its first two nonzero far-field terms.

(d) Differentiate your approximate potential and compare the result with your field expansion. How many terms in the potential must be retained to reproduce the requested field terms?

(e) Introduce the distance $r = x - d/2$ from the midpoint of the charges. Expand the field through the term proportional to d^3/r^5. Explain why there is no term proportional to d^2/r^4.

7. A particle of mass m moves along a horizontal line. At $t = 0$, its position is zero and its velocity is $v_0 > 0$. The only horizontal force is

$$F = -bv,$$

where $b > 0$ is constant.

(a) Write Newton's second law and solve for $v(t)$.

(b) Find $x(t)$. Determine the limiting position as $t \to \infty$.

(c) Does the particle come to rest at any finite time? Is its total distance traveled finite?

(d) When the speed has fallen to v_0/n, where $n > 1$, what fraction of the total distance has been traveled?

(e) Eliminate time to obtain v as a function of x. Obtain the same result directly using $a = v\,dv/dx$.

(f) Find the total work done by the resistive force. Explain how a particle that never stops at a finite time can nevertheless lose all its initial kinetic energy in the limit.

8. A straight rod rotates in a horizontal plane with constant angular velocity $\omega > 0$ about one end O. A small bead is driven outward along the rod at a constant speed $u > 0$ relative to the rod. At $t = 0$, the bead is at O, and the rod lies along the positive x axis.

(a) Find $x(t)$ and $y(t)$. Eliminate time to obtain the trajectory in polar coordinates.

(b) Differentiate your Cartesian position functions to find the velocity. Show that the speed is

$$v = u\sqrt{1 + \omega^2 t^2}.$$

(c) Find the acceleration vector. Resolve it into components parallel and perpendicular to the rod.

(d) Find the tangential acceleration dv/dt. Is this equal to the magnitude of the acceleration? Explain.

(e) Find the normal acceleration and hence the radius of curvature of the trajectory as a function of time.

(f) If the bead has mass m, find the net power delivered to it. Check your result by differentiating its kinetic energy.

9. The total energy of a relativistic particle of mass m and momentum magnitude p is

$$E = \sqrt{m^2c^4 + p^2c^2}.$$

(a) For $p \ll mc$, expand E through the term proportional to p^4. Identify the rest energy and the leading approximation to the kinetic energy.

(b) Find the first correction to the Newtonian kinetic energy $p^2/(2m)$. Is the Newtonian expression an overestimate or an underestimate at fixed momentum?

(c) Given that the particle's speed satisfies $v = dE/dp$, derive the exact relation between v and p. Expand this relation through the term proportional to p^3.

(d) Estimate the value of $p/(mc)$ at which the magnitude of the leading correction to the Newtonian kinetic energy is 1% of the Newtonian term.

(e) In the opposite limit $p \gg mc$, find the first two terms in the expansion of E. Does the speed approach c from above or below?

10. A particle of mass m initially moves with speed $v_0 > 0$ along a straight line. While it is moving, a resistive force of magnitude bv^n acts opposite to its velocity, where $b > 0$ and $n \geq 0$. Once it reaches rest, it remains at rest.

(a) Find the dimensions of b. Use dimensional analysis to construct a characteristic time and a characteristic length from m, b, and v_0. Explain why these need not equal the actual stopping time and stopping distance.

(b) Solve

$$m\frac{dv}{dt} = -bv^n$$

for $n \neq 1$, and solve the case $n = 1$ separately. State the time interval on which each solution describes forward motion.

(c) For which values of n does the particle reach rest in a finite time?

(d) Using $a = v\,dv/dx$, determine for which values of n the total distance traveled is finite. Find that distance when it exists.

(e) Describe the long-time behavior for $n = 1$, $n = 2$, and $n = 3$. In particular, distinguish approaching zero speed from approaching a finite limiting position.

(f) For $n > 2$, the characteristic length constructed in part (a) decreases as v_0 increases. Explain why this does not imply that a faster particle travels a smaller total distance.

11. A thin hemispherical shell of radius R and mass M rests with its circular rim on a horizontal table, forming a dome. Water of density ρ is introduced slowly through a small opening at the highest point. The opening remains exposed to atmospheric pressure.

Assume that the rim forms a watertight seal until the shell begins to lift, and neglect surface tension. Atmospheric pressure acts on the exterior of the shell.

(a) When the dome is just full, find the hydrostatic pressure at height z above the table, where $0 \le z \le R$.

(b) Divide the curved surface into thin horizontal rings. Find the upward component of the pressure force on a ring of height dz, and integrate to obtain the total upward force on the shell.

(c) Find the mass M for which the shell is just about to lift when it becomes full.

(d) Obtain the same result by considering the shell and enclosed water as one system. Include the pressure force exerted by the table on the water.

(e) Before the dome is full, let the water surface be at height $h < R$. The air above the water is at atmospheric pressure. Find the upward pressure force on the shell as a function of h.

(f) Determine the range of shell masses for which lifting occurs before the water reaches the top.